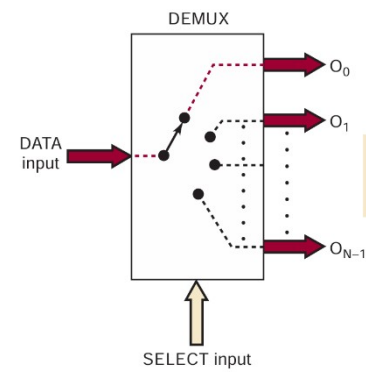
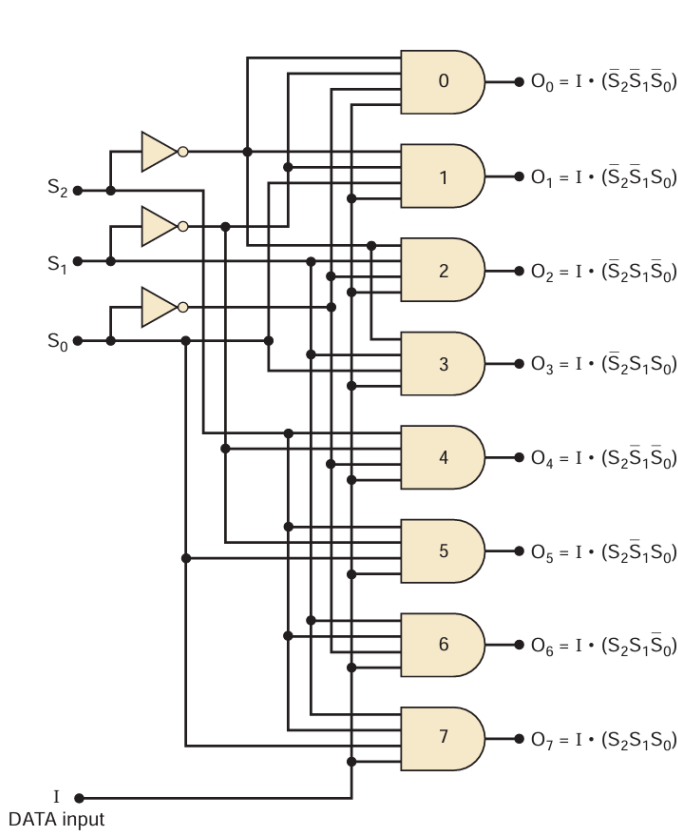


8×1 Line Demultiplexer



SELECT code			OUTPUTS							
S ₂	S ₁	S ₀	O ₇	O ₆	O ₅	O ₄	O ₃	O ₂	O ₁	O ₀
0	0	0	0	0	0	0	0	0	0	I
0	0	1	0	0	0	0	0	0	I	0
0	1	0	0	0	0	0	0	I	0	0
0	1	1	0	0	0	0	I	0	0	0
1	0	0	0	0	0	I	0	0	0	0
1	0	1	0	0	I	0	0	0	0	0
1	1	0	0	I	0	0	0	0	0	0
1	1	1	I	0	0	0	0	0	0	0

Verilog code with test-bench with Truth table:

```

module demux_1x8 (
input wire i,
input wire [2:0] sel,
output reg [7:0] out);

always @(*) begin
    out = 8'b00000000;
    case (sel)
        3'b000: out[0] = i;
        3'b001: out[1] = i;
        3'b010: out[2] = i;
        3'b011: out[3] = i;
        3'b100: out[4] = i;
        3'b101: out[5] = i;
        3'b110: out[6] = i;
        3'b111: out[7] = i;
    endcase
end

endmodule

```

```

`timescale 1ns/1ps
module demux_1x8_tb;

reg i;
reg [2:0] sel;
wire [7:0] out;

demux_1x8 one (
    .i(i),
    .sel(sel),
    .out(out)
);

integer k;

initial begin

    $display("sel      out");
    $display("-----");

    i = 1'b1;
    for (k = 0; k < 8; k = k + 1) begin
        sel = k;
        #10;
        $display(" %b      %b", sel, out);
    end

    $finish;
end

endmodule

```

sel	out
000	00000001
001	00000010
010	00000100
011	00001000
100	00010000
101	00100000
110	01000000
111	10000000

Verilog code with test-bench with random Inputs:

```
`timescale 1ns/1ps

module demux_1x8_tb;

reg i;
reg [2:0] sel;
wire [7:0] out;

demux_1x8 dut (
    .i(i),
    .sel(sel),
    .out(out)
);

integer k;

initial begin

    // Generate waveform
    $dumpfile("demux_1x8.vcd");
    $dumpvars(0, demux_1x8_tb);

    $display(" Random Select Tests");
    $display("-----");
    $display(" i      sel      out");
    $display("-----");
    repeat (10) begin
        sel = $random % 8;
        i = $random % 2;
        #10;
        $display(" %b      %b      %b", i, sel, out);
    end

    $finish;
end

endmodule
```

Random Select Tests		
i	sel	out
1	100	00010000
1	001	00000010
1	101	00100000
0	101	00000000
1	001	00000010
1	110	01000000
0	101	00000000
0	001	00000000
0	101	00000000
1	101	00100000

